

SATHYAGITH VIJAYKUMAR

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Education

Indian Institute of Technology Bhubaneswar

Bachelor of Technology in Mechanical Engineering

2023 – Present

CGPA: 8.42/10

Sri Chaitanya Techno School (CBSE)

12th Standard

2022

89.6%

Ryan International School (CBSE)

10th Standard

2020

95.8%

Work Experience

Generative Products Inc.

Remote, United States

AI Technical Intern – Data Intelligence Platform & Autonomous Customer Support | [\[GitHub\]](#)

May 2026 – July 2026

- Spearheaded backend design, development and cloud deployment of an enterprise-grade conversational AI platform for CRM, covering Data Intelligence and autonomous customer support, on AWS EC2 with Docker.
- Owned end-to-end development and deployment of a LangGraph-orchestrated **Data Intelligence Platform**, enabling agentic analytics for enterprise clients; delivered **Structured AutoML** (profiling, SHAP, MLOps), **Enterprise GPT** as a custom knowledge base with three retrieval modes—structured SQL, document RAG and Neo4j graph RAG with citations—connected to operational connectors such as OMS and CRM, **Metabot** NL-to-SQL analytics with interactive charts and root-cause analysis and a scientific assistant with grounded answers, follow-up suggestions and a **LiveKit** voice agent for spoken operational and research queries.
- Built and deployed an **autonomous customer support platform** from the ground up with omnichannel intake, RAG-based answers, automated ticketing, Jira escalation, LangGraph workflows and LangFuse-based LLM evaluation.
- Implemented a **Customer Digital Twin** (A2A multi-agent) for hyper-personalized dialogue over live traffic, using TOD-BERT for intent recognition, **GOPO** for dynamic strategy selection (LinUCB, ESND CG rewards), **MetaRank** for personalized offer ranking and NeMo Guardrails and Presidio for policy and PII enforcement.

Projects

Route Resilience – Occlusion-Robust Road Extraction & Graph Criticality | [\[GitHub\]](#)

Jun 2026 – Jul 2026

- Developed an end-to-end Earth Observation pipeline to assess and quantify **urban road-network resilience** from satellite imagery under occlusion.
- Implemented **DeepLabV3+ and UNet++** for occlusion-robust road segmentation (validation IoU: **0.569**) and used SAM for mask repair without retraining.
- Engineered graph reconstruction via skeletonization and MST/Union-Find, identifying critical **gatekeeper junctions** through betweenness centrality and quantifying resilience via targeted node-removal experiments; visualized results on a React/Leaflet dashboard.

Krishi Drone (KD) – Autonomous Precision Agriculture Drone | [\[GitHub\]](#)

Oct 2025 – Mar 2026

- Engineered an autonomous agricultural drone for intelligent **crop monitoring and targeted pest treatment**, combining computer vision with autonomous flight.
- Used **OpenCV** to detect infected plants and enable selective biopesticide spraying on affected crop areas.
- Implemented autonomous navigation and control with **ROS2, PID/LQR and waypoint planning**, validated in simulation and on hardware.

Semantra.io – AI-Powered Knowledge Graph Assistant | [\[GitHub\]](#)

Jul 2025 – Aug 2025

- Built a multi-agent assistant for **semantic and geospatial information retrieval** across public data portals including MOSDAC and VEDAS.
- Designed a modular agentic system for document ingestion, retrieval and **knowledge-graph construction** using RAG, Neo4j, PostGIS and Google A2A.
- Enabled natural-language dialogue and **spatial reasoning** through context-aware retrieval over heterogeneous geospatial datasets.

Context-Aware Geospatial Data Retrieval Using LLM/NLP | [\[GitHub\]](#)

Jul 2024 – Aug 2024

- Developed an NLP-driven system for **natural-language discovery of geospatial datasets** with intuitive query and retrieval workflows.
- Combined geo-parsing, voice search and context-aware retrieval using **Apache Airflow, PostgreSQL (PostGIS), Apache Feast and Streamlit** to simplify access to spatial data.

ROS Hexapod Control via EEG and Speech Recognition | [\[GitHub-EEG\]](#) [\[GitHub-Speech\]](#)

Jun 2024 – Aug 2024

- Developed a multimodal human-robot interface for real-time hexapod control using **BCI (EEG) and speech commands** in ROS.
- Deployed EEG-based navigation on **NVIDIA Jetson Nano** and integrated speech-to-command control for hands-free robot operation.

Technical Skills

- **Languages:** C, C++, Python
- **Robotics:** ROS2, Gazebo, Nav2, MoveIt2, sensor fusion, PID/LQR control
- **Databases & Search:** PostgreSQL (PostGIS, pgvector, Apache AGE), Neo4j, OpenSearch, DuckDB
- **Frameworks & Libraries:** FastAPI, Apache Airflow, PyTorch, LangGraph, A2A, LangFuse, LiveKit, River, NetworkX, OpenCV
- **Cloud & DevOps:** AWS EC2, Docker
- **AI & ML:** Machine Learning, Deep Learning, Reinforcement Learning, Transformers, NLP, RAG, LLM evaluation, semantic segmentation, graph analytics, agentic workflows

Achievements

- Selected for **Stage 2**, the penultimate round of the **e-Yantra Robotics Competition (IIT Bombay)**, Agricultural Drone theme (Top 50 among 2000+ teams).
- Secured **9th position** at the **Inter-IIT 13th Tech Meet – Idea Forge** (December 2024), a product ideation track for a drone-related product concept.
- Participated in multiple **Bharatiya Antariksh Hackathons (ISRO)** across 2024–2026.